1. Organic Chemistry

1.1. Formulae, Functional Group and Terminology

- Organic Chemistry is the scientific study of the structure, properties, and reactions of organic carbon compounds.

Suffix	Compound type
-ane	Alkane
-ene	Alkene
-anol	Alcohol
-anoic acid	Carboxylic acid
-yl/-anoate	Ester

Homologous Series: a group of organic compounds that have similar chemical properties due to being part of the:

- Same Functional Group
- Same General Formula
- Differing from One member to the next by a CH_2 unit
- Displaying a trend in physical properties
- Sharing similar chemical properties

Essential Terminology

1. Functional Group: an **atom or group of atoms** that **determine the chemical properties** of a homologous series
2. Structural Isomers: compounds with the **same molecular formula** but **different structural formula.**
3. Saturated Compounds: molecules in which all carbon-carbon bonds are **single bonds.**
4. Unsaturated Compounds: molecules where **one or more carbon-carbon bonds** are not single.

1.2. Fuels

Common fossil fuels include: coal, natural gas [main constituent: methane] and petroleum

Petroleum: a mixture of hydrocarbons (**Carbon and Hydrogen ONLY**) which can be separated into useful fractions by fractional distillation

Going down the fractions

- Increasing chain length
- Increasing temperature
- Lower volatility
- Increase Boiling Points
- Increase Viscosity (harder to flow)

Petroleum Fraction	Uses
Refinery gas	heating and cooking
Gasoline fraction	fuels for cars
Naphtha fraction	chemical feedstock
Kerosene/Paraffin fraction	jet fuel
Diesel oil/Gas oil fraction	fuel in diesel engines
Fuel oil fraction	fuel in ships and home heating systems
Lubricating fraction	lubricants, waxes and polishes
Bitumen	making roads

Complete & Incomplete Combustion

Complete Combustion

A complete combustion reaction with any organic compounds will produce **Carbon Dioxide and Water** as its product. The general equation is:

$C_XH_Y + O_2 \rightarrow CO_2 + H_2O$

Incomplete Combustion

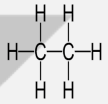
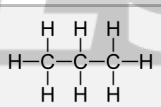
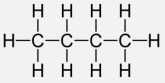
An incomplete combustion reaction with any organic compounds will produce **Carbon Monoxide and Water** as its product. The general equation is:

$C_XH_Y + O_2 \rightarrow CO + H_2O$

1.3. Alkanes

Alkanes: saturated hydrocarbons [single carbon bonds] that are generally unreactive; however, they do undergo combustion reactions

General formula = C_nH_{2n+2}

Methane: CH_4 (n=1)	Ethane: C_2H_6 (n=2)
	
Propane: C_3H_8 (n=3)	**Butane: C_4H_{10} (n=4)**
	

Substitution Reaction

Alkanes go through substitutional reactions, where the atom of another element under UV light replaces the hydrogen atom. (Photochemical)

- The presence of UV light is to **provide activation energy**

$CH_4 + Cl_2 \rightarrow$ (light) $\rightarrow HCl + CH_3Cl / CH_2Cl_2 / CHCl_3 / CCl_4$

Compounds = chloromethane/di/tri/tetrachloromethane

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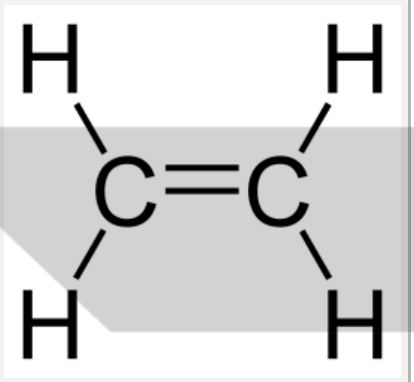
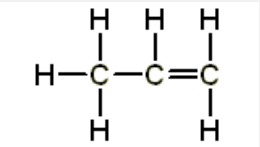
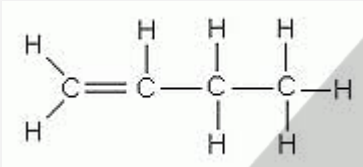
1.4. Alkenes

Alkenes: unsaturated hydrocarbons [at least one double bond between 2 carbon atoms]

Have isomers: same molecular formula but different structural formula (placement of double bond shifts), e.g. but-1-ene and but-2-ene

General formula = C_nH_{2n}

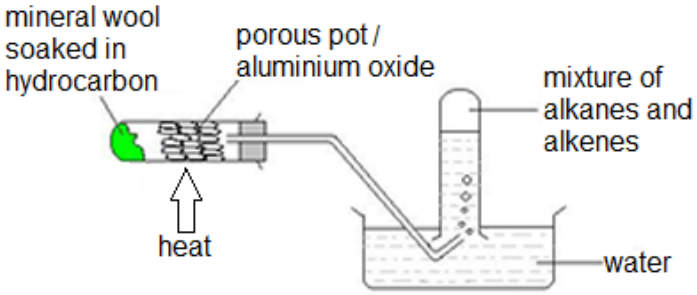
Functional group: C=C bond

(n=1)	Ethene:C_2H_4 (n=2)
N/A	
Propene:C_3H_6 (n=3)	But-1-ene:C_4H_8 (n=4)
	

Catalytic Cracking (Alkane)

- Thermal decomposition reaction at 700 Celsius, in which an alkene (and sometimes hydrogen) is produced from an alkane. It also form shorter-chain alkanes.
- Hydrocarbon is heated, and vapours are passed over a catalyst (alumina (Al_2O_3 and silica)
- **Cracking** always produces a short-chain compound with a C=C bond **(matches the supply of fractions with demand and produces alkene for feedstock.)**

  e.g. Cracking of ethane will give ethene and hydrogen



- Butane → Ethane + Ethene ; $C_4H_{10} \rightarrow C_2H_6 + C_2H_4$

How do we distinguish between saturated and unsaturated hydrocarbons?

Using Bromine water (reddish-brown):

- Saturated: remains reddish brown (unreactive)
- Unsaturated: turns colourless/decolourises

1.7. Alkenes' Addition Reactions

Alkene addition reactions **only form one product.**

1. **With Bromine:** (the test for saturation - orange brown-colourless)

   e.g. ethene (g) + bromine (aq) → 1,2-dibromoethane (l)
2. **With steam (hydration):** forms alcohols with heat (300°C), pressure (**60atm/6000kPa**) and an acid catalyst (phosphoric acid)

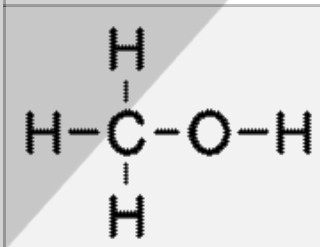
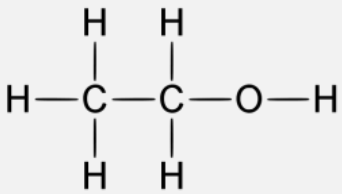
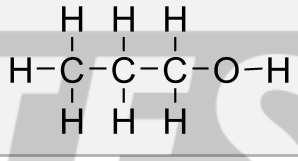
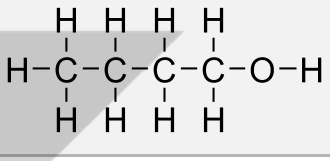
   e.g. ethene (g) + steam (g) ⇌ ethanol (l)
3. **With hydrogen (hydrogenation),** a double bond breaks down to form an alkane with a heat of 200 degrees and a catalyst (nickel)

   e.g. ethene (g) + hydrogen (g) → ethane (g)

1.8. Alcohols

General formula = $C_nH_{2n+1}OH$

Functional group: OH

Methanol: CH_3OH (n=1)	Ethanol: C_2H_5OH (n=2)
	
Propanol: C_3H_7OH (n=3)	Butanol: C_4H_9OH (n=4)
	

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Methods of Production

Fermentation of Aqueous Glucose (for Ethanol)	Catalytic Addition of Steam to Ethene
Yeast is added to dissolved glucose. The products ethanol, carbon dioxide, and the temperature between 25-35 °C are used for optimal enzyme activity.	The ethene reacts with steam (reversibly) to form ethanol in the following conditions: **300°C, 60 atm (6000 kPa).** Catalyst - phosphoric acid [while low temp gives better yield, high temp is used for a faster rate of reaction]
The slow reaction produces a dilute solution that requires processing. Can only be produced in batches	The fast reaction produces pure ethanol. Continuous production (no batches)
Produces greenhouse gas (CO_2)	No greenhouse pollutants
Uses renewable resources	Uses non-renewable resources (crude oil)

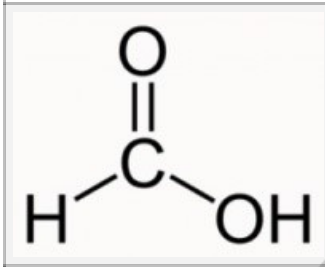
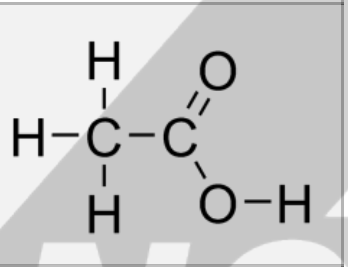
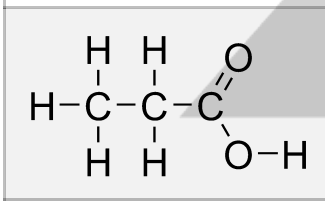
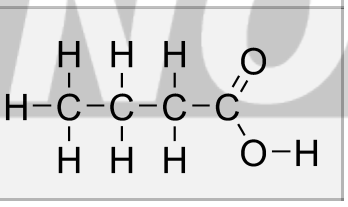
Uses of Ethanol:

- Solvent in glues, printing inks & perfumes
- Fuel

1.9. Carboxylic Acids

General formula: $C_nH_{2n+1}COOH$

Functional group: COOH

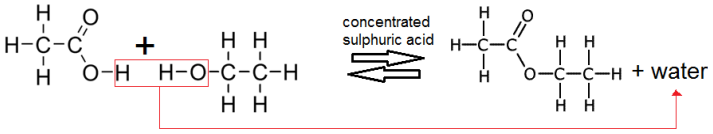
Methanoic Acid: CH_2O_2 (n=0)	Ethanoic Acid: CH_3COOH (n=1)
	
Propanoic Acid: $C_3H_6O_2$ (n=2)	Butanoic acid: $C_4H_8O_2$ (n=3)
	

Properties of Ethanoic Acid

- Weak acid with high pH and low dissociation
- Formed by:
  - Bacterial Oxidation of Ethanol (Vinegar Production)
  - With acidified aqueous potassium manganate (VII) (Good Oxidising Agent)

Carboxylic acids react with alcohols (with an acid catalyst/Heat) to give esters in a **condensation reaction**, for example:

- Ethanoic acid + ethanol ⇌ ethyl ethanoate + water (alcohol = -yl & carboxylic acid = -anoate)



- Carboxylic Acids also have **different structural isomers.**

Reactions with Carboxylic Acid

- Magnesium reacts with ethanoic acid to form magnesium ethanoate and hydrogen gas.
  - $2CH_3COOH + Mg \rightarrow (CH_3COO)_2Mg + H_2$
- When propanoic acid reacts with hydroxides, it will form salt and water (Neutralisation reaction)
  - $CH_3CH_2COOH + KOH \rightarrow CH_3CH_2COOK + H_2O$
- When butanoic acid reacts with carbonates, it will form salt, water, and carbon dioxide.
  - $2CH_3CH_2CH_2COOH + K_2CO3 \rightarrow 2CH_3CH_2CH_2COOK + H_2O + CO_2$

1.11. Polymers

- Large molecules are built up from small units known as monomers.
- Monomers are **a single unit of a substance.**
- Polymers can have different linkages depending on the type of polymerisation and monomer

Example:

Small units (monomers)	Linkages
Polyester (PET)	Ester
Polyamide & Protein	Amide

Plastics (PET)

Plastics (**polyethylene terephthalate**) are made from polymers. However, they are not biodegradable. (NOTE: PET can be hydrolysed back to monomers and re-polymerised)

Environment Challenges caused by Plastics

1. Disposal in Landfill sites
2. Accumulation in Oceans
3. Formations of toxic gases from burning

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1.12. Properties of Polymers

- It can be moulded under heat and pressure due to its low density
- Low conductivity (retain heat)
- Resistant to corrosion
- Non-biodegradable and lead to plastic waste

1.13. Addition and Condensation Polymerisation

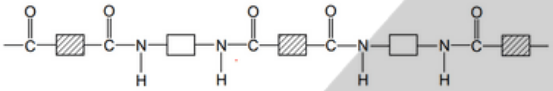
Addition Polymerisation

- This only occurs in monomers that contain double carbon (**C=C**) bonds.
- Polymers produced using alkene monomers
- Forms only a polymer molecule
- Poly(ethene): is a polymer produced from ethene by addition polymerisation

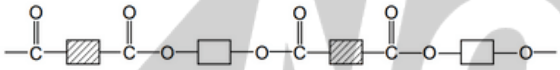
Double bond splits, and the polymer is formed. (Polymers have no double bonds)

Condensation Polymerisation

- When two different monomers are linked together with the removal of a smaller molecule, usually **water** (forms one H_20 molecules per linkage).
    1. **Nylon** (polyamide) is made from a **dicarboxylic acid monomer and a diamine monomer** (a compound with an NH_2 functional group). Forms **amide linkage.**



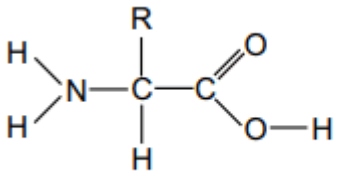
    2. **PET** (polyester) is made from a dicarboxylic acid monomer and diols (alcohol with an -OH functional group). Forms **ester linkage.**



1.14. Protein & Natural Polyamides

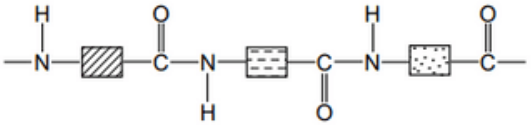
Proteins act as the natural polyamides formed from amino acids monomers.

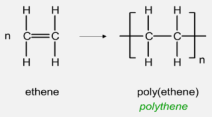
General Structure of Amino Acids:



Proteins:

- Proteins contain the same linkages (amide links) as nylon but with different units from amino acid monomers. Their main structure you should know is:



	Use	Structure
Poly(ethene)	Plastic bags and gloves, clingfilm (low density), mugs, bowls, chairs, dustbins (high density)	

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